MADASOURCE

Complete Platform Architecture & Functional Requirements

B2B International Trade & Digital Trade Facilitation Platform
Initial Market: Madagascar → International Markets
Product Categories: Vanilla · Cloves · Cocoa

1. PLATFORM VISION

MadaSource is not intended to be only a product marketplace.

The platform should progressively become a:

B2B Marketplace
↓
→ **Supplier Discovery Platform**
↓
→ **RFQ & Matching Platform**
↓
→ **Quotation & Negotiation Platform**
↓
→ **Trade Management Platform**
↓
→ **Digital Trade Facilitation Platform**
↓
→ **Digital Trade Infrastructure for Madagascar**

The fundamental objective is: Allow an international buyer to discover a Malagasy exporter, request a quotation, compare offers, negotiate, confirm a transaction and monitor the entire trade journey from one centralized platform.

2. CORE BUSINESS MODEL

The initial transaction architecture is:

INTERNATIONAL BUYER
↓
MADASOURCE PLATFORM
↓
MALAGASY EXPORTER

The platform must support two possible commercial models.

Model A — MadaSource as Commercial Reseller / Trader

Exporter provides a contractual net price. MadaSource adds its commercial margin.

Exporter net price: USD 40/kg
MadaSource commercial price: USD 43/kg
Buyer sees: USD 43/kg

The system must internally maintain: supplier net price; MadaSource commercial margin; final buyer price.

Model B — MadaSource as Intermediary

MadaSource can act as an intermediary and receive a contractually defined service/commission fee.

The architecture must support both models. The exact legal, tax and contractual structure will be defined separately.

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3. PLATFORM MODULES

The complete architecture should contain the following major modules:

- A. Public Website
- B. Authentication & Identity
- C. Buyer Portal
- D. Exporter Portal
- E. Product Marketplace
- F. RFQ Engine
- G. Supplier Matching Engine
- H. Quotation Engine
- I. Negotiation Engine
- J. Trade Management Engine
- K. Trade Journey / Timeline
- L. Document Management
- M. Sample Request Management
- N. Internal Messaging
- O. Notification System
- P. Payment Management
- Q. Logistics & Shipment Tracking
- R. Supplier Verification
- S. Reputation & Performance
- T. MadaSource AI Assistant
- U. Administration Console
- V. Partner Management
- W. Reporting & Analytics
- X. Audit & Compliance

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4. PUBLIC WEBSITE

The public website should include:

- Home
- About MadaSource
- Products
- Exporters
- How MadaSource Works
- Buyer Registration
- Exporter Registration
- Request a Quote
- Contact
- Terms & Conditions
- Privacy Policy
- Legal / Compliance

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5. PRODUCT MARKETPLACE

Initial categories: **VANILLA · CLOVES · COCOA**

The marketplace must support:

- Product search
- Category filtering
- Country/origin
- Quality
- Grade
- Certification
- Available quantity
- MOQ
- Packaging
- Incoterm
- Exporter
- Verification status
- Price indication

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6. PRODUCT DATA STRUCTURE

Each product listing should contain:

General

- Product name
- Product category
- Description
- Origin
- Region of origin
- Product specifications
- Images
- Documents

Commercial

- Available quantity
- Unit
- MOQ
- Price indication
- Currency
- Incoterm
- Payment terms
- Quote validity

Quality

- Grade
- Quality parameters
- Moisture
- Size
- Purity
- Processing type
- Certification

The system must allow product-specific attributes because vanilla, cloves and cocoa do not necessarily have identical specifications.

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7. EXPORTER PROFILE

Each exporter must have a professional company profile. Information:

- Company name
- Logo
- Description
- Location
- Contact information
- Business registration information
- Years in business
- Export experience
- Export markets
- Products
- Available capacity
- Certifications
- Verification status
- Performance indicators
- Reviews

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8. MADASOURCE VERIFICATION SYSTEM

Create a structured verification system.

LEVEL 1 — PROFILE VERIFIED

Basic company information verified.

LEVEL 2 — BUSINESS VERIFIED

Business registration and company documentation reviewed.

LEVEL 3 — EXPORT VERIFIED

Business verified + export capability and relevant documentation reviewed.

LEVEL 4 — MADASOURCE PREMIUM VERIFIED

Additional verification, documents, inspections or other due diligence where applicable.

The system must record: verification level; verification date; expiry date where applicable; documents reviewed; administrator responsible; verification notes; status.

Possible statuses: Pending · Verified · Rejected · Expired · Suspended

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9. DIGITAL EXPORTER PASSPORT

Every verified exporter should have a public MadaSource profile URL, e.g. *madasource.com/exporter/company-name*.

Generate a unique QR code for this profile. The QR code should redirect directly to the company's MadaSource profile, creating a **MADASOURCE DIGITAL EXPORTER PASSPORT**.

The exporter can use the QR code on:

- Business cards
- Brochures
- Catalogues
- Exhibition booths
- Packaging
- Commercial documents
- Promotional materials

The objective is for MadaSource to become an independent trust/reference layer for Malagasy exporters.

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10. BUYER REGISTRATION

Buyer registration fields:

- Company name
- Country
- Region
- Business type
- Website
- Contact person
- Email
- Phone
- Business address
- Import markets
- Products of interest

Buyer account statuses: Pending · Active · Suspended · Blocked

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11. EXPORTER REGISTRATION

Exporter onboarding fields:

- Company name
- Contact person
- Email
- Phone
- Address
- Region
- Business registration
- Tax information
- Export documentation
- Export experience
- Products
- Capacity
- Certifications
- Bank/payment information

Sensitive documents must not be publicly visible.

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12. RFQ — REQUEST FOR QUOTATION ENGINE

This is one of the most important modules. Buyer creates an RFQ with fields:

- Product
- Quantity
- Unit
- Quality
- Grade
- Certification
- Packaging
- Destination country
- Destination port
- Incoterm
- Target delivery date
- Target price
- Payment preference
- Additional requirements

RFQ status flow:

Draft
↓
Submitted
↓
Matching
↓
Sent to Exporters
↓
Quotations Received
↓
Negotiating
↓
Accepted
↓
Closed

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13. NATURAL LANGUAGE RFQ

The buyer should also be able to enter a request such as:

> “I need 20 tons of Grade A Madagascar vanilla CIF Hamburg in October.”

The platform should convert the request into structured fields, for example:

Product: Vanilla
Quantity: 20,000 kg
Grade: A
Destination: Hamburg
Incoterm: CIF
Delivery: October

The system should identify missing information and ask the buyer to complete it.

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14. RFQ MATCHING ENGINE

After an RFQ is submitted, MadaSource should identify suitable exporters. Matching criteria:

- Product
- Available quantity
- Capacity
- Quality
- Grade

- Certification
- Destination
- Incoterm
- Exporter verification
- Price
- Lead time
- Historical performance
- Response rate

```
Exporter A — 94% Match
Exporter B — 89% Match
Exporter C — 83% Match
```

The score must be based on actual platform data and configurable weighting rules. **Do not generate arbitrary/random scores.**

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15. EXPORTER RFQ INBOX

Exporters receive matching RFQs, for example:

```
NEW RFQ
Buyer: Germany
Product: Vanilla
Quantity: 20,000 KG
Quality: Grade A
Certification: Organic
Destination: Hamburg
Incoterm: CIF
Expected Delivery: October 2026
```

Actions: Respond · Decline · Ask Question

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16. QUOTATION ENGINE

Exporter creates a quotation. Fields:

- Quote ID
- Product
- Quantity
- Unit price
- Currency
- Quality
- Grade

- Certification
- Packaging
- Incoterm
- Destination
- Lead time
- Payment terms
- Validity
- Additional conditions

QUOTE MSQ-2026-000145
Product: Madagascar Vanilla
Quantity: 20,000 kg
Price: USD 40/kg
Incoterm: FOB
Validity: 7 days

17. COMMERCIAL PRICE ENGINE

The platform should support a price calculation engine, for example:

Supplier Price: USD 40/kg
Logistics: USD 1.20/kg
Insurance: USD 0.20/kg
Other applicable costs: USD 0.30/kg
MadaSource Margin: USD 3.30/kg
= Final Commercial Price: USD 45/kg

The architecture must allow the calculation rules to be configured by MadaSource administrators.

18. QUOTE VALIDITY

Every quotation and external cost should have: creation date; last update; expiry date; source; status.

Possible status: Active · Expiring · Expired

The system must prevent expired quotations from being presented as current offers.

19. QUOTE COMPARISON

Buyer can compare multiple quotations, for example:

Exporter	Price	Quality	MOQ	Certification	Incoterm	Lead Time
Exporter A	$42/kg	Grade A	5T	Organic	FOB	15 days

Exporter B	$44/kg	Grade A	10T	Organic	CIF	20 days
Exporter C	$41/kg	Grade B	5T	Standard	FOB	12 days

Actions: View Offer · Negotiate · Accept

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20. NEGOTIATION ENGINE

Buyer can submit a counter offer, for example:

```
Exporter: USD 45/kg
Buyer: USD 42/kg
Exporter: USD 44/kg
Buyer: ACCEPT
```

Every negotiation event must be stored. No negotiation history should be deleted.

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21. NEGOTIATION HISTORY

Store: original quotation; counteroffers; timestamps; users; prices; quantities; conditions; messages; final accepted terms.

This creates an auditable commercial history.

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22. TRADE ID

Once a quotation is accepted, automatically create a unique **MADASOURCE TRADE ID**, e.g. *MST-2026-000145*.

The Trade ID becomes the central reference for the entire transaction.

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23. TRADE RECORD

Each trade contains:

- Trade ID
- Buyer
- Exporter
- Product
- Quantity
- Quality
- Price
- Currency

- Incoterm
- Destination
- Payment terms
- Delivery target
- Documents
- Messages
- Payment status
- Shipment status
- Trade status
- Event history

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24. TRADE STATE MACHINE

TRADE CONFIRMED
↓
PAYMENT PENDING
↓
PAYMENT CONFIRMED
↓
PREPARATION
↓
READY FOR SHIPMENT
↓
SHIPPED
↓
IN TRANSIT
↓
DELIVERED
↓
COMPLETED

Exceptions: Cancelled · Disputed · On Hold

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25. TRADE JOURNEY

Create a visual timeline, for example:

RFQ Submitted
↓

Exporter Matched
↓
Quotation Received
↓
Negotiation
↓
Trade Confirmed
↓
Payment
↓
Preparation
↓
Inspection
↓
Shipment
↓
In Transit
↓
Delivered
↓
Completed

Each event must store: date; time; actor; action; comment; supporting document where applicable.

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26. DOCUMENT MANAGEMENT SYSTEM

Create a centralized document vault. Documents can be associated with: company; RFQ; quotation; trade; shipment.

Document categories

- Commercial Documents
- Company Documents
- Quality Documents
- Export Documents
- Shipping Documents
- Payment Documents
- Certification Documents

Supported formats: PDF · JPG · PNG · DOCX

Store: document name; type; owner; upload date; version; status; related Trade ID.

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27. SAMPLE REQUEST SYSTEM

Buyer can request a sample before committing to a large order. Workflow:

Request Sample
↓
Sample Quotation
↓
Sample Confirmed
↓
Sample Payment
↓
Sample Shipped
↓
Sample Received
↓
Sample Approved / Rejected

A sample request can later be connected to an RFQ.

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28. INTERNAL MESSAGING

Create secure platform messaging. Participants: Buyer ↔ Exporter.

Messages can be attached to: RFQ; quotation; negotiation; trade; sample request.

The system should preserve conversation history.

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29. NOTIFICATION SYSTEM

Notifications should be generated automatically, for example:

- “New RFQ received.”
- “New quotation received.”
- “Buyer submitted a counteroffer.”
- “Quotation accepted.”
- “New Trade created.”
- “Payment confirmed.”
- “New document uploaded.”
- “Trade status updated.”
- “Sample shipped.”

Notifications should be available: in-app; email-ready architecture; push-ready architecture.

30. PAYMENT MANAGEMENT

For the first implementation, create payment management architecture without requiring direct banking integration.

Payment statuses: Pending · Processing · Confirmed · Failed · Refunded

Allow: bank transfer record; payment reference; payment voucher upload; payment date; amount; currency; admin confirmation.

Future architecture must support: payment gateways; banking partners; letters of credit; escrow-like mechanisms where legally available; financing partners.

MadaSource itself must not be architected as a bank.

31. LOGISTICS MANAGEMENT

Initial version: allow manual shipment information. Fields:

- Shipping method
- Carrier
- Shipment reference
- Container number
- Tracking number
- Origin
- Destination
- Estimated departure
- Estimated arrival

Statuses: Pending · Ready for Shipment · Shipped · In Transit · Arrived · Delivered

Future versions should support API integrations with logistics providers.

32. CIF / FOB / CFR PRICE ARCHITECTURE

The system should support Incoterms. Initial: FOB · CFR · CIF. Architecture must allow future Incoterms.

For CIF calculations: Product price + Freight + Insurance = CIF base cost. Then MadaSource commercial pricing rules may be applied according to the commercial model.

The system must clearly distinguish: Supplier Cost · External Costs · MadaSource Margin · Final Buyer Price.

33. COST & QUOTATION SOURCES

Every externally provided price should store: source; provider; date; validity; currency; unit; destination; Incoterm; confirmation status.

Freight: USD 3,200
Source: Logistics Partner
Updated: 09/08/2026
Valid: 7 days
Status: Active → EXPIRED (when applicable)

The platform must not silently use outdated data.

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34. ADMIN DASHBOARD

The administration console is central. Admin modules:

- Dashboard
- Buyers
- Exporters
- Products
- RFQs
- Matching
- Quotations
- Negotiations
- Trades
- Samples
- Documents
- Payments
- Shipments
- Verification
- Messages
- Reviews
- Partners
- Reports
- Settings
- Audit Logs

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35. ADMIN — BUYER MANAGEMENT

Admin can:

- View buyers
- Approve/suspend accounts
- View company information
- View RFQs
- View quotations
- View trades
- View disputes
- Manage account status

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36. ADMIN — EXPORTER MANAGEMENT

Admin can:

- Review applications
- Review documents
- Verify businesses
- Assign verification level
- Approve
- Reject
- Suspend
- Reactivate
- View products
- View RFQs
- View quotations
- View trades
- View performance

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37. ADMIN — RFQ MANAGEMENT

Admin can see: all RFQs; status; buyer; product; destination; matching exporters; quotations received; RFQ age; unresolved RFQs.

Admin should be able to intervene manually when required.

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38. ADMIN — QUOTATION MANAGEMENT

Admin can see: supplier quotation; price; validity; negotiation history; final status.

Admin may intervene for exceptional/high-value transactions.

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39. ADMIN — TRADE MANAGEMENT

Admin can see all active trades on a dashboard with: Trade ID, Buyer, Exporter, Product, Quantity, Value, Payment, Documents, Shipment, Current Status.

Admin should have an overview of every active transaction.

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40. EXCEPTION MANAGEMENT

Create an explicit **EXCEPTION CENTER**. Examples:

- Payment delayed
- Quotation expired
- Exporter not responding
- Document missing
- Shipment delayed
- Buyer dispute
- Quality issue
- Quantity discrepancy
- Cancelled order

Admin can: investigate; assign; comment; change status; resolve; escalate.

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41. DISPUTE MANAGEMENT

Create a basic dispute module. Buyer or exporter can open a dispute with fields: Trade ID; Reason; Description; Evidence; Documents; Requested resolution.

Statuses: Open · Under Review · Resolution Proposed · Resolved · Closed

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42. REPUTATION SYSTEM

After completed trades: buyer can rate exporter, exporter can rate buyer.

Metrics: communication; product conformity; delivery; documentation; responsiveness; transaction experience.

Do not generate artificial reviews or ratings. New companies should display “NEW ON MADASOURCE” until sufficient transaction history exists.

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43. EXPORTER PERFORMANCE

Calculate from real data:

- Response rate
- Average response time
- Quotation acceptance rate
- Completed trades
- Cancellation rate
- Dispute rate
- Buyer rating

These metrics should contribute to future matching.

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44. BUYER PERFORMANCE

Similarly prepare: completed trades; payment reliability; cancellation rate; dispute rate; exporter rating.

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45. MADASOURCE AI TRADE ASSISTANT

Create an AI layer called the **MADASOURCE AI TRADE ASSISTANT**. The assistant should help users navigate the platform.

Buyer example

User: “I need 20 tons of Madagascar vanilla CIF Dubai.”
AI: Identify missing information, then create structured RFQ.

Exporter example

Exporter: “I have 10 tons of Grade A vanilla available.”
AI: Help create a product listing.

Trade example

Buyer: “Why did my CIF price increase?”
AI should analyze actual stored cost data and explain the difference.

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46. AI SAFETY PRINCIPLE

AI must never invent:

- Suppliers

- Prices
- Capacity
- Certifications
- Documents
- Logistics status
- Payments
- Transactions

If information is unavailable: **INFORMATION NOT AVAILABLE**.

The AI must distinguish between: Verified Data · Platform Data · Estimated Data · User Input · AI Recommendation.

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47. AI MATCHING

The AI may assist the matching engine but must not override deterministic business rules without authorization.

Matching should combine: Structured Rules · Historical Data · AI Assistance.

This is important for explainability and reliability.

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48. BUYER DASHBOARD

- Overview
- My RFQs
- Quotations
- Negotiations
- Active Trades
- Completed Trades
- Sample Requests
- Saved Exporters
- Saved Products
- Documents
- Messages
- Notifications
- Company Profile
- Settings

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49. EXPORTER DASHBOARD

- Overview
- RFQ Inbox
- Quotations
- Negotiations
- Active Orders / Trades
- Products
- Capacity
- Documents
- Company Profile
- Verification
- Messages
- Notifications
- Performance

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50. SEARCH & DISCOVERY

Global search must support: Product · Exporter · RFQ · Trade — according to user permissions.

Buyer-facing search should prioritize: Product · Exporter · Quality · Certification · Availability · Price · Verification.

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51. FAVORITES

Buyer can save: products; exporters; quotations.

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52. AUDIT LOG

Every important action must generate an audit event, for example:

- User created RFQ
- Exporter submitted quotation
- Buyer counteroffered
- Exporter accepted
- Trade created
- Payment confirmed

- Document uploaded
- Trade status changed
- Admin verified exporter

Audit records should contain: actor; action; object; timestamp; previous value; new value where applicable.

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53. ROLE-BASED ACCESS CONTROL

Minimum roles: Buyer · Exporter · MadaSource Admin

Architecture should support future: Operations Admin · Compliance Admin · Finance Admin · Logistics Admin · Super Admin · Partner

Permissions must be role-based.

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54. MULTI-TENANT COMPANY ARCHITECTURE

Do not design the system only around individual users. Use: Organization → Users → Roles → Permissions.

A company may eventually have several employees, for example:

```
Company: ABC Export
Users: Director, Sales Manager, Operations Manager
```

All users belong to the same organization but have different permissions.

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55. DATABASE CORE ENTITIES

Minimum database entities, grouped by domain:

Domain	Entities
Identity	Users, Organizations, Roles, Permissions
Marketplace	Products, ProductCategories, ProductListings, ProductSpecifications
Exporters	Exporters, ExporterProfiles, VerificationRecords, Certifications, CapacityRecords
Buyers	BuyerProfiles
RFQ	RFQs, RFQItems, RFQMatches
Commercial	Quotes, QuoteItems, Negotiations, CounterOffers
Trade	Trades, TradeItems, TradeEvents, TradeStatus
Logistics	Shipments, TrackingEvents
Payment	Payments, PaymentRecords, PaymentDocuments
Documents	Documents, DocumentVersions
Communication	Conversations, Messages, Notifications

Samples	SampleRequests, SampleShipments
Reputation	Reviews, Ratings, PerformanceMetrics
Administration	AuditLogs, Disputes, Exceptions
Partners	Partners, PartnerServices, PartnerConfigurations

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56. API-FIRST ARCHITECTURE

The platform should be designed with APIs. Core API domains:

/auth · /users · /organizations · /products · /exporters · /buyers · /rfqs · /matches · /quotes · /negotiations · /trades · /documents · /samples · /messages · /notifications · /payments · /shipments · /reviews · /admin · /analytics

This is essential because future integrations will depend on APIs.

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57. FUTURE PARTNER INTEGRATION ARCHITECTURE

The platform should be prepared for:

- Banking APIs
- Payment Gateway APIs
- Logistics APIs
- Shipment Tracking APIs
- Inspection APIs
- Certification APIs
- Customs/Trade Data APIs
- Currency/FX APIs

These should be implemented through modular integration layers.

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58. INTERNATIONALIZATION

Initial languages: English · French | **Future:** Malagasy

The system must support localization without rebuilding the frontend.

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59. CURRENCY

Initial: USD · EUR | **Future:** MGA, GBP, AED, CNY, etc.

Currency conversion must be separated from core transaction data. Historical transaction prices must never change because an exchange rate changes.

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60. SECURITY

- Encrypted passwords
- Secure authentication
- Authorization
- Role-based access
- Protected API endpoints
- Input validation
- File validation
- Secure document storage
- Session security
- Audit logging
- Rate limiting
- Database security
- Backup strategy

Sensitive company documents and financial information must never be publicly accessible.

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61. DATA PRIVACY

Public profile data must be separated from private business information, for example:

Public	Private
Company Name, Products, Verification, General Location, Certifications	Bank information, Internal documents, Contracts, Private communications, Commercial margins, Sensitive business info

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62. MOBILE RESPONSIVENESS

The platform must work on: Desktop · Tablet · Mobile. Buyer and exporter workflows should be fully usable on mobile.

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63. ADMIN ANALYTICS

Dashboard should include:

- Total Buyers
- Total Exporters
- Verified Exporters
- Active RFQs
- Open RFQs
- Quotes Received
- Active Trades
- Completed Trades
- Trade Value
- MadaSource Gross Commercial Margin
- Average RFQ Response Time
- Top Products
- Top Destinations
- Exporter Performance
- Unresolved Exceptions

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64. TRADE INTELLIGENCE

The architecture should allow MadaSource to eventually understand:

- Which products are requested
- Which countries buy
- Average price
- Requested quantities
- Demand trends
- Exporter capacity
- Response rates
- Recurring buyers
- Seasonal demand

This data should become a strategic asset of the platform.

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65. REPEAT ORDER ENGINE

After a completed transaction, the system should be able to suggest a **REORDER**, for example:

Buyer purchased: 20 tons Vanilla
Date: March 2026
Six months later: “Your next purchasing cycle may be approaching. Would you like to request a new quotation?”

The platform can prepare a new RFQ based on historical data.

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66. TRADE ID AS CENTRAL SYSTEM REFERENCE

Every major object should be linkable to the Trade ID, for example:

RFQ: RFQ-2026-0012
Quote: MSQ-2026-0044
Trade: MST-2026-000145
Documents → linked to MST-2026-000145
Messages → linked to MST-2026-000145
Payment → linked to MST-2026-000145
Shipment → linked to MST-2026-000145

This creates a single digital transaction record.

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67. END-TO-END TRANSACTION EXAMPLE

The system must support this complete journey:

Step 1 — International buyer registers.

↓

Step 2 — Buyer searches: Madagascar Vanilla

↓

Step 3 — Buyer compares verified exporters.

↓

Step 4 — Buyer creates: 20,000 KG Vanilla RFQ

↓

Step 5 — MadaSource matching engine identifies compatible exporters.

↓

Step 6 — Exporters receive RFQ.

↓

Step 7 — Exporter submits quotation.

↓

Step 8 — Buyer compares quotations.

↓

Step 9 — Buyer submits counteroffer.

↓

Step 10 — Exporter accepts.

↓

Step 11 — MadaSource generates: MST-2026-000145

↓

Step 12 — Trade dashboard becomes active.

↓

Step 13 — Payment is recorded.

↓

Step 14 — Documents are uploaded.

↓

Step 15 — Exporter updates preparation status.

↓

Step 16 — Shipment information is added.

↓

Step 17 — Shipment becomes: IN TRANSIT

↓

Step 18 — Buyer receives goods.

↓

Step 19 — Trade becomes: COMPLETED

↓

Step 20 — Buyer and exporter can rate each other.

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68. DEMO ENVIRONMENT

Because certain external services may not be integrated initially, the platform should provide a clearly labelled **DEMO MODE**.

Demo Mode can simulate: payment confirmation; shipment status; delivery; partner confirmation.

However, the interface and database workflow must be the same as the future real system. **Never display simulated transactions as real transactions.**

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69. V1 IMPLEMENTATION PRIORITY

Although the architecture must be complete, development should be modular.

Phase 1 — Core

Authentication, Buyer, Exporter, Marketplace, Products, RFQ, Matching, Quotation, Negotiation, Trade, Trade ID, Trade Dashboard.

Phase 2 — Operations

Documents, Messaging, Notifications, Sample Requests, Verification, Reviews, Exception Management.

Phase 3 — Integrations

Payments, Logistics, Shipment Tracking, Inspection, Certification, Banking.

Phase 4 — Intelligence

AI Assistant, Advanced Matching, Trade Analytics, Demand Intelligence, Price Intelligence, Predictive Tools.

70. IMPORTANT DEVELOPMENT PRINCIPLE

Do not build a monolithic system where every future feature is hard-coded into the first version. Use:

- Modular architecture
- API-first design
- Role-based access
- Configurable workflows
- Feature flags
- Extensible database model
- Integration layer

This will allow MadaSource to expand progressively.

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71. FEATURE FLAGS

Prepare feature flags such as: VANILLA_ENABLED, CLOVES_ENABLED, COCOA_ENABLED.

Future: PRODUCER_MODULE_ENABLED, COLLECTOR_MODULE_ENABLED, COOPERATIVE_MODULE_ENABLED, BANKING_MODULE_ENABLED, LOGISTICS_MODULE_ENABLED, INSPECTION_MODULE_ENABLED, INSURANCE_MODULE_ENABLED, CUSTOMS_MODULE_ENABLED.

The future modules should remain disabled until MadaSource decides to activate them.

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72. CRITICAL UX PRINCIPLE

The user should not need to understand the complexity of international trade. MadaSource should hide unnecessary complexity.

For example, instead of requiring the buyer to manually contact the Exporter, Freight Forwarder, Insurance, Inspection, Bank and Customs separately, the platform should progressively orchestrate these processes through one **MADASOURCE TRADE JOURNEY**.

The complexity should exist inside the platform, not in the user's experience.

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73. CORE DIFFERENTIATOR

MadaSource must not be developed as simply “Alibaba for Madagascar.” The platform should be designed to become:

> **The digital trade infrastructure that makes Madagascar easier to buy from.**

The marketplace is only the entry point. The real value is the orchestration of the transaction.

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74. SUCCESS CRITERIA

The most important acceptance test is:

> Can an international buyer and a Malagasy exporter complete a structured B2B trade journey through MadaSource without leaving the platform for the core workflow?

The platform should successfully support:

DISCOVER
↓
RFQ
↓
MATCH
↓
QUOTE
↓
NEGOTIATE
↓
ACCEPT
↓
TRADE ID
↓
DOCUMENTS
↓
PAYMENT RECORD
↓
SHIPMENT
↓
TRACKING
↓
DELIVERY
↓
COMPLETION

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75. IMPORTANT LEGAL / REGULATORY NOTE

The technical architecture should not assume that MadaSource itself is legally authorized to perform regulated activities.

The system must support integration with licensed/authorized third parties where required, in particular:

- Banking
- Payment services

- Insurance
- Customs brokerage
- Freight forwarding
- Inspection
- Certification
- Regulated export activities

The final legal structure, contractual responsibilities, tax treatment, licensing requirements and compliance requirements will be determined separately for Madagascar and the relevant destination markets.

The software architecture must therefore be legally adaptable rather than hard-coded around one assumed legal structure.

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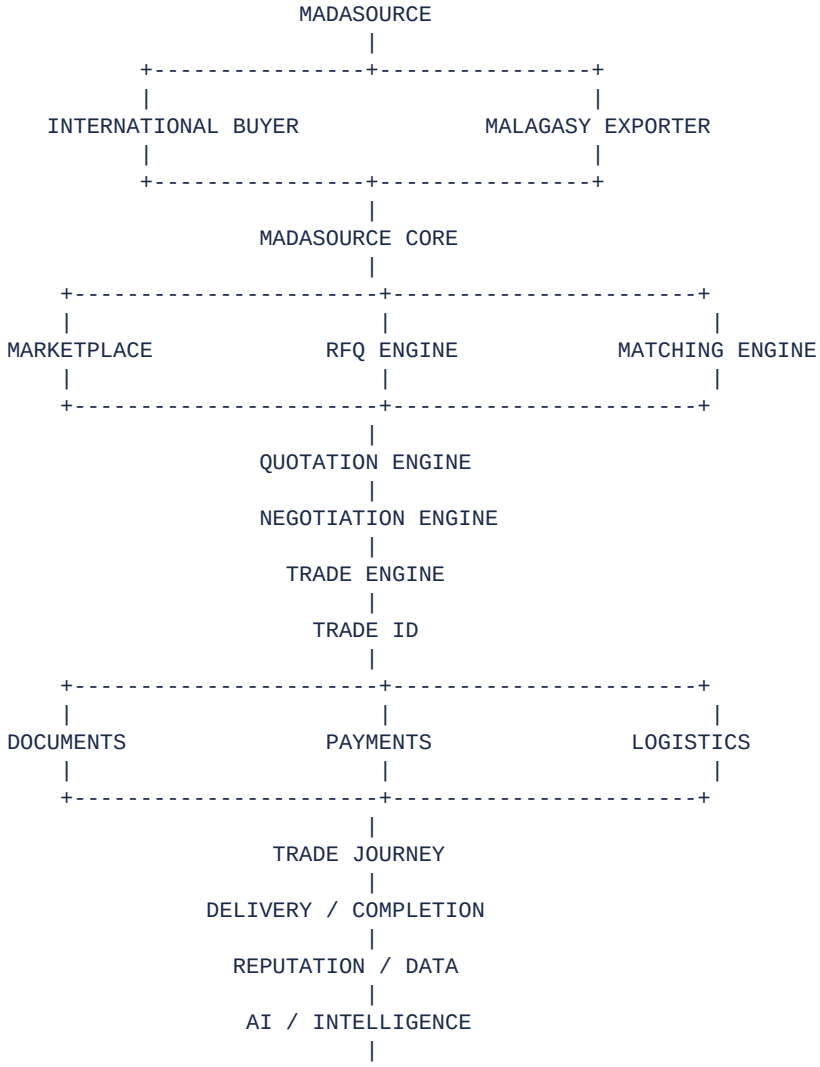
76. FINAL PRODUCT ARCHITECTURE

The complete conceptual architecture is:



FUTURE PARTNER ECOSYSTEM

77. REQUEST TO THE DEVELOPMENT COMPANY

Based on this architecture, please provide:

- 1. Technical architecture
- 2. Recommended technology stack
- 3. Frontend architecture
- 4. Backend architecture
- 5. Database architecture
- 6. API architecture
- 7. Cloud / hosting architecture
- 8. Security architecture
- 9. File/document storage architecture
- 10. Authentication architecture
- 11. Role & permission architecture
- 12. Estimated development timeline
- 13. Development cost
- 14. Maintenance cost
- 15. Source-code ownership
- 16. Deployment process
- 17. Database ownership
- 18. API documentation
- 19. Testing strategy
- 20. Backup and disaster recovery
- 21. Scalability strategy
- 22. Future integration strategy

Please also clearly separate:

- A. Features included in the initial development
- B. Features prepared architecturally but not activated
- C. Third-party integrations required later
- D. Features requiring additional development cost

FINAL REQUIREMENT

Please do not interpret this document as a request to build all future modules immediately. The objective is to create a strong and scalable technical foundation.

The first operational scope is strictly:

Scope	Detail
3 Products	Vanilla + Cloves + Cocoa
2 Core User Types	International Buyers + Malagasy Exporters
1 Core Journey	RFQ → Matching → Quotation → Negotiation → Trade → Documents → Payment Record → Shipment → D

The architecture must nevertheless allow MadaSource to progressively evolve into a much larger digital trade infrastructure without requiring a complete rewrite of the platform.